# Raisa Nusrat

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## Research Interests

Deep learning for medical imaging; facial analysis for neurological disorder detection (stroke, ALS, Parkinson's disease); multimodal fusion of imaging and clinical data; explainable and generalizable healthcare AI; bio-inspired robotics and autonomous drones.

## Education

**B.Sc. in Electrical and Electronic Engineering** 2020 – June 2026
Bangladesh University of Engineering and Technology (BUET), Dhaka CGPA: 3.28/4.00
Thesis: *A Multimodal Deep Learning Framework for Stroke Detection from Facial Asymmetry and Clinical Data with Explainable AI and External Validation* — Supervisor: Dr. Mohammad Faisal, Professor, EEE, BUET

## Publications

**Journal Articles**

1. **R. Nusrat**, M. R. Rahman, I. J. Riya, and M. Faisal, "Age-Robust Stroke Detection and Paretic-Side Identification Using Nasolabial Fold Asymmetry and Clinical Data," *IEEE Journal of Biomedical and Health Informatics*, under review, 2026.
2. **R. Nusrat**, M. S. Raihan, S. M. R. Rahman, H. Saha, and S. S. Ahbab, "A Low-Cost Integrated Motorcycle Theft Detection and Monitoring System Using Arduino, ESP32-CAM, and GSM/GPS Technologies," *Journal of the Electrical and Electronic Engineering Division, Institution of Engineers, Bangladesh (IEB)*, accepted, 2026.

**Conference Papers**

3. **R. Nusrat**, "When Scale Flips the Answer: Corridor-Level Deep Reinforcement Learning for Traffic Signal Control on a Simulated Dhaka Arterial," *AABEA 2026 Biennial Convention*; abstract accepted for proceedings, full paper under review.
4. **R. Nusrat**, "Rare-Attack Blindness in Federated Intrusion Detection for the Internet of Medical Things," *14th International Conference on Electrical and Computer Engineering (ICECE 2026)*, Dhaka, under review.

**Manuscripts in Preparation**

5. **R. Nusrat** et al., "Differentiating stroke from ALS using facial-motion asymmetry on the Toronto NeuroFace dataset."
6. **R. Nusrat** et al., "Parkinson's disease detection using the PADS smartwatch dataset."
7. **R. Nusrat**, A. Hamim, M. Fahim, M. J. Hasan, and A. Hasan, "HIVE-R: Bio-inspired and visual odometry based drone for rescue operations in GPS-denied environments."

## Research Experience

**Undergraduate Thesis Research — Stroke Detection** 2025 – 2026
*Department of EEE, BUET | Supervisor: Dr. Mohammad Faisal*

- Built a clinical dataset of 117 patients (71 stroke, 46 non-stroke) from hospital records, with facial images and 38 laboratory features.
- Designed a Siamese CNN (EfficientNet-B0) comparing left and right facial regions; combined image AUC of 0.983.
- Fused image and clinical models; 80% accuracy and 88.9% sensitivity on 15 held-out external patients.

**Clinical Study — Multimodal Stroke Detection (Facial Imaging and Blood Biomarkers)** 2026 – Present

- Wrote the study protocol and obtained BUET Ethical Review Committee approval (Ref. BUET/ERC/2026/01).
- Supported by the Basic Research Grant, RISE, BUET.

**Independent Research — Deep Reinforcement Learning for Traffic Signal Control** 2026

- Built a five-junction Dhaka corridor in SUMO and trained a shared-policy DQN; under saturation, cut arterial delay by 60% and cross-street delay by 41% versus fixed-time control.

**Robotics Research — HIVE-R Autonomous Rescue Drone** 2025 – Present

- Honeybee-inspired drone swarm for search and rescue in GPS-denied environments.
- Camera-IMU visual odometry (SLAM) on Pixhawk and Raspberry Pi; YOLOv8-Nano human detection with fuzzy augmentation for smoke, blur, and low light.

## Selected Projects

**Multi-Modal Parkinson's Disease Detection** *EEE 402, AI & ML Lab*
Detection from speech, gait, and handwriting with late fusion.

**Motorcycle Theft Detection and Monitoring System** *Embedded Systems*
Arduino and ESP32-CAM unit with PIN-locked ignition, live camera view, and GPS/SMS tracking; working prototype at about 3800 BDT.

**Autonomous Trash-Collecting Line-Following Robot** *EEE 318, Control Systems Lab*
PID-controlled line follower with object detection and collection.

**Noise Cancellation Using Adaptive Filters** *EEE 312, DSP Lab*
Compared LMS and RLS adaptive filters by SNR and mean squared error.

**Microgrid Simulation with Intentional Islanding** *EEE 306, Power Systems Lab*
Simulink model of a neighbourhood microgrid that islands from the grid during faults.

## Teaching Experience

**Mathematics Tutor (International)** 2026 – Present
Teaching students in Australia on the Edexcel curriculum; personalized lesson plans and progress tracking.

**Private Tutor (International)** 2024 – 2025
One-on-one online tutoring for a Stony Brook University (USA) undergraduate in circuit analysis, signals and systems, and advanced calculus.

**Content Developer and Academic Support, Udvash Academic & Admission Care** 2022 – 2023
Wrote conceptual questions and parallel texts; provided real-time online Q&A support.

**Private Tutor (National Curriculum)** 2022 – Present
Physics and Mathematics for Classes 9–12 (SSC/HSC).

## Industrial Training

**Industrial Attachment, Bangladesh Television (BTV)**
Satellite communication, high-power RF transmission, and studio broadcasting electronics; hands-on signal acquisition and hardware operation.

## Awards and Activities

- Champion, Integration Bee, organized by Tachyon.
- Participant, Robotics Workshop.

## Technical Skills

**Programming:** Python, MATLAB, C/C++, LaTeX
**Machine Learning:** PyTorch, scikit-learn, OpenCV, YOLOv8
**Design and Simulation:** MATLAB/Simulink, SUMO, PSpice, Proteus, Verilog, Quartus, AutoCAD
**Hardware:** Arduino, ESP32-CAM, Raspberry Pi, Pixhawk (MAVLink), IMU, GPS/GSM modules

## Test Scores

TOEFL iBT: scheduled 14 November 2026